

Name

KEY

GT ID

Evans

CHEM 2312 LS

Organic Chemistry II

Midterm Examination 2

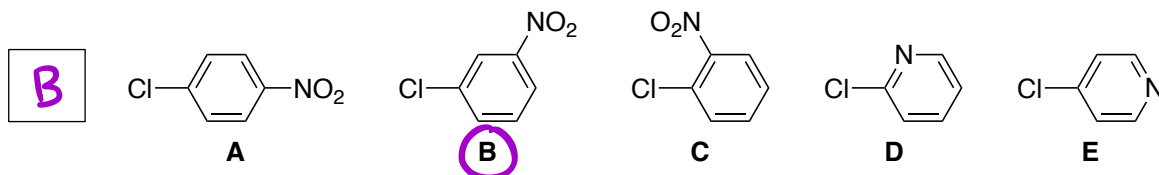
Take a deep breath. You got this.

Do not open this exam until you are instructed to do so. Only a pencil or pen, scratch paper, and molecular model kit are permitted. You will have 50 minutes to complete the exam; it can be submitted at any time after 25 minutes have elapsed. Write all responses clearly in the boxes or spaces provided; work outside of designated areas will not be graded. On multiple-choice questions, write the letter of your choice in the box *and* circle your choice.

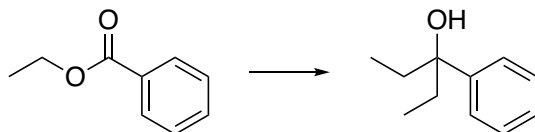
Ensure that configurations can be unambiguously assigned at all stereocenters (*R/S* or *E/Z*). Use “+ En” to indicate the presence of a racemic mixture when chiral products are generated from achiral starting materials.

Multiple-choice Questions. Select the best response. Circle your choice and write its letter in the box. Each question is worth **4 points**.

1. Which of the following compounds reacts most *slowly* with ammonia (NH_3)?

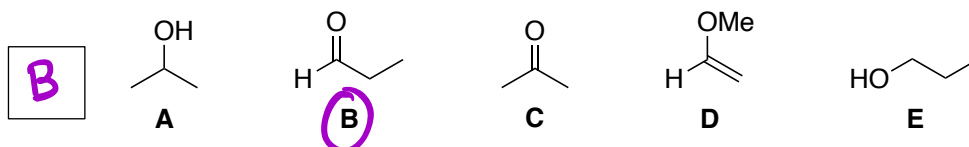
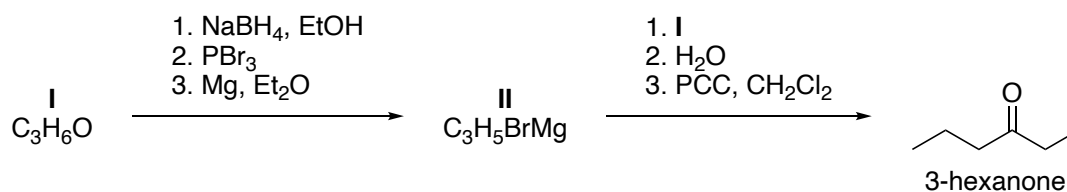


2. What set of reagents would accomplish the transformation below?

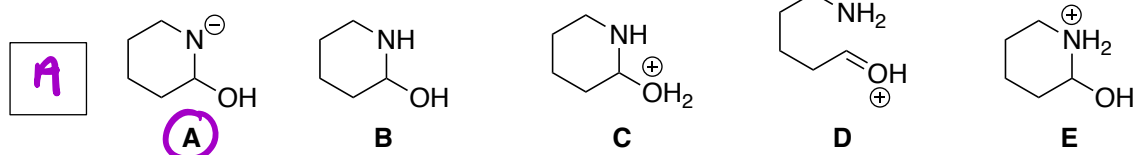
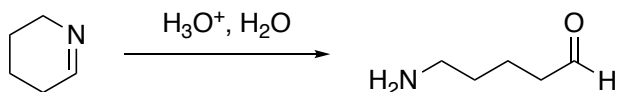


- C
- A. 1. LiAlH_4 ; 2. H_2O
 - B. 1. NaOEt ; 2. H_3O^+
 - C. 1. $\text{CH}_3\text{CH}_2\text{MgBr}$; 2. H_2O
 - D. 1. LDA ; 2. $\text{CH}_3\text{CH}_2\text{Br}$
 - E. NaBH_4 , EtOH

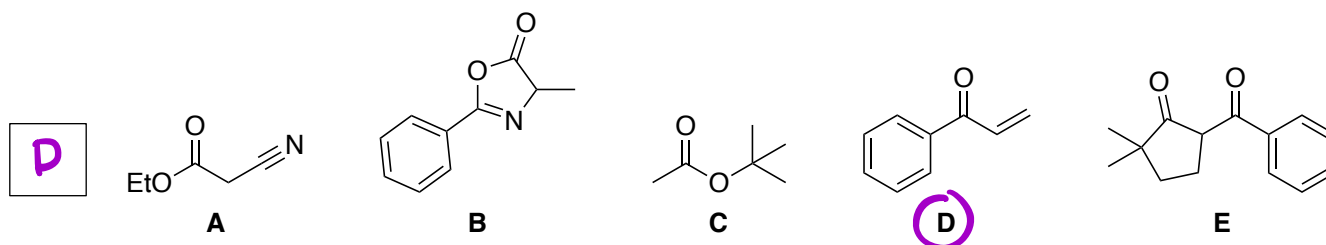
3. Elaboration of compound **I** to Grignard reagent **II**, followed by treatment of **II** with **I** and reaction with pyridinium chlorochromate (PCC), gave 3-hexanone. What is the structure of compound **I**?



4. Which of the following structures is *least* likely to be a reactive intermediate in the mechanism of the reaction below?

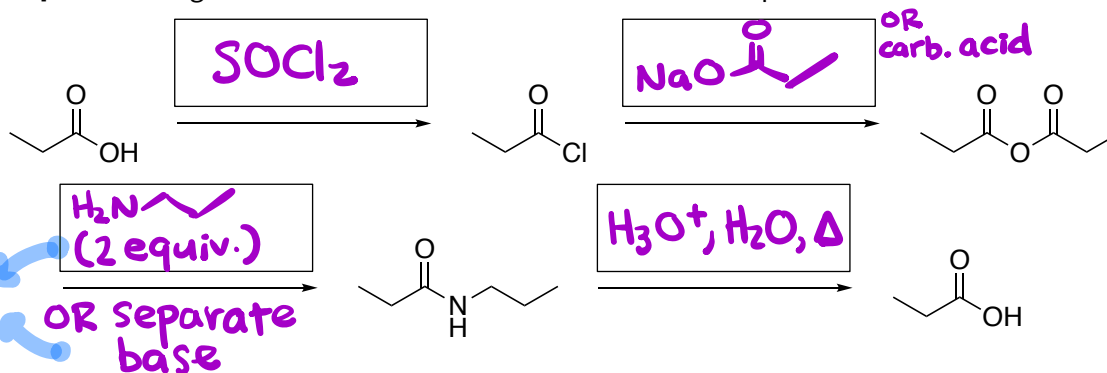


5. Which of the following compounds *cannot* form an enol?



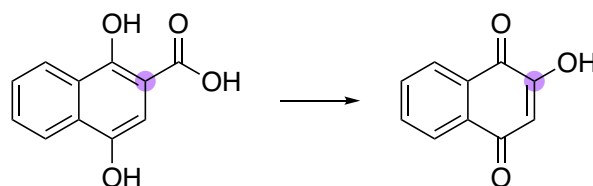
Other Questions. Place your response in the designated space for each question.

6. **12 pts.** Add reagents above each arrow to show how to accomplish each transformation.

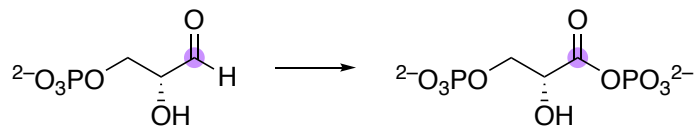


7. **4 pts.** Indicate whether oxidation, reduction, or no change in oxidation state is occurring at the highlighted carbon in each transformation below.

A. ☒ Oxidation ☐ Reduction ☐ No change

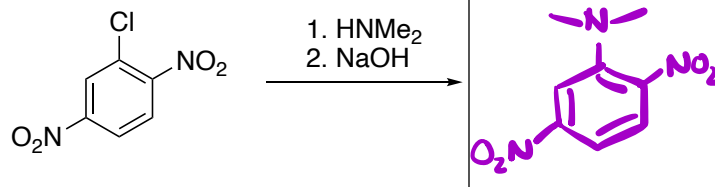


B. ☒ Oxidation ☐ Reduction ☐ No change

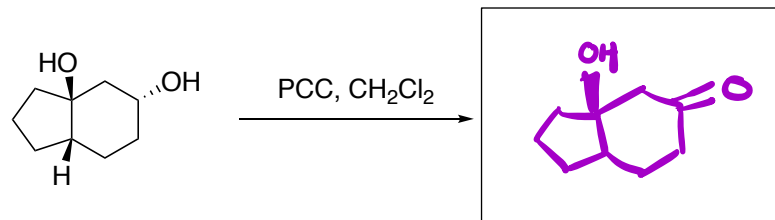


8. **32 pts.** Predict the major organic product of each reaction or sequence of reactions below. If multiple isomers are possible, draw only the major product.

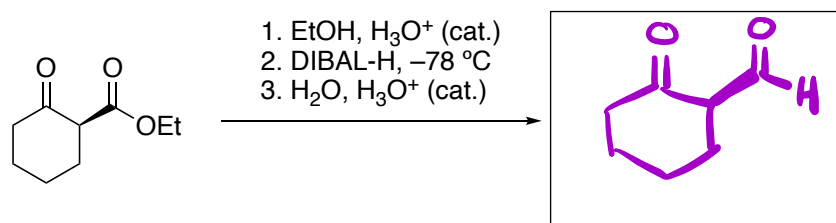
A.



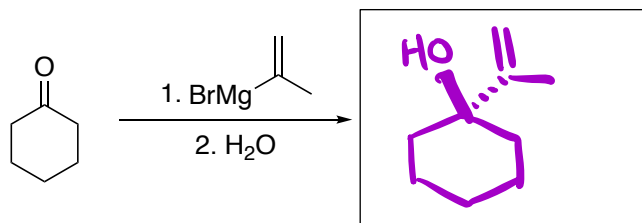
B.



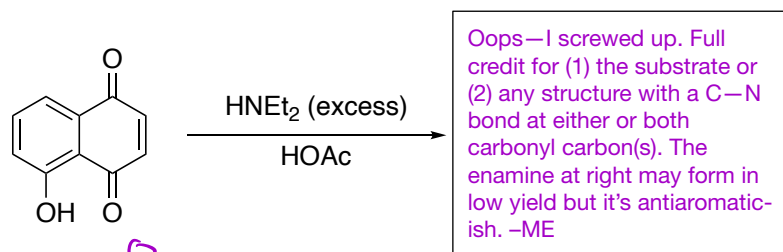
C.



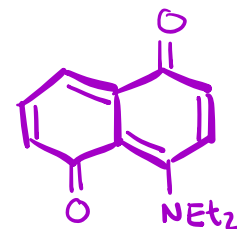
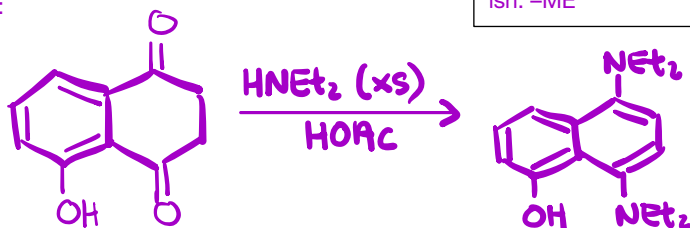
D.



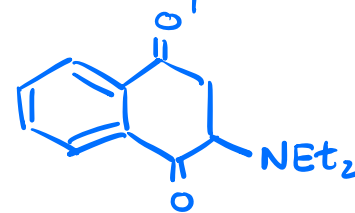
E.



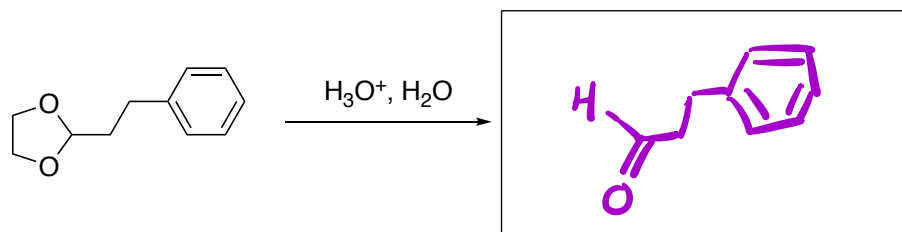
The reaction should've been:



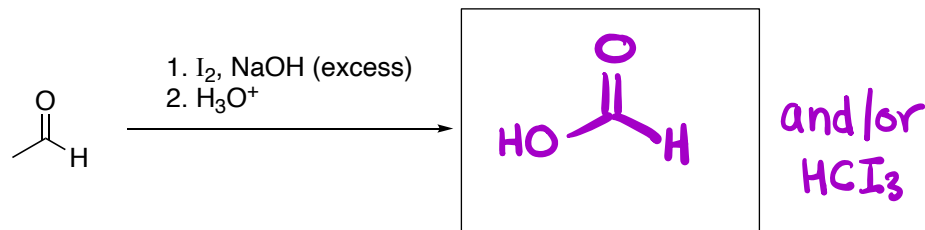
Conjugate add'n.
also okay:



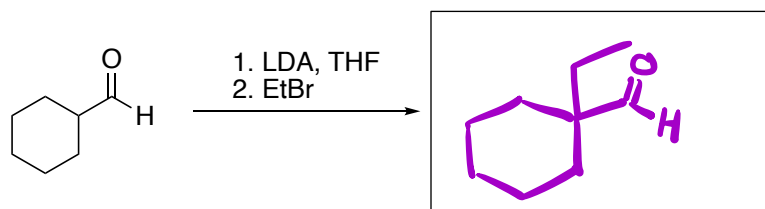
F.



G.

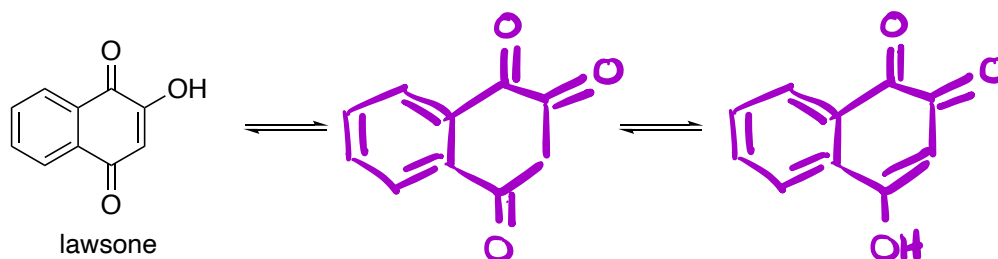


H.

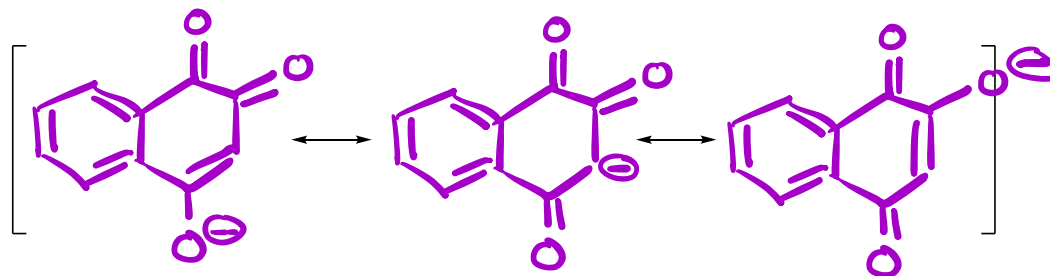


9. **10 pts.** Lawsone is a red-orange dye present in the leaves of the henna plant.

A. Draw two tautomers of lawsone, the structure of which is given at left below.

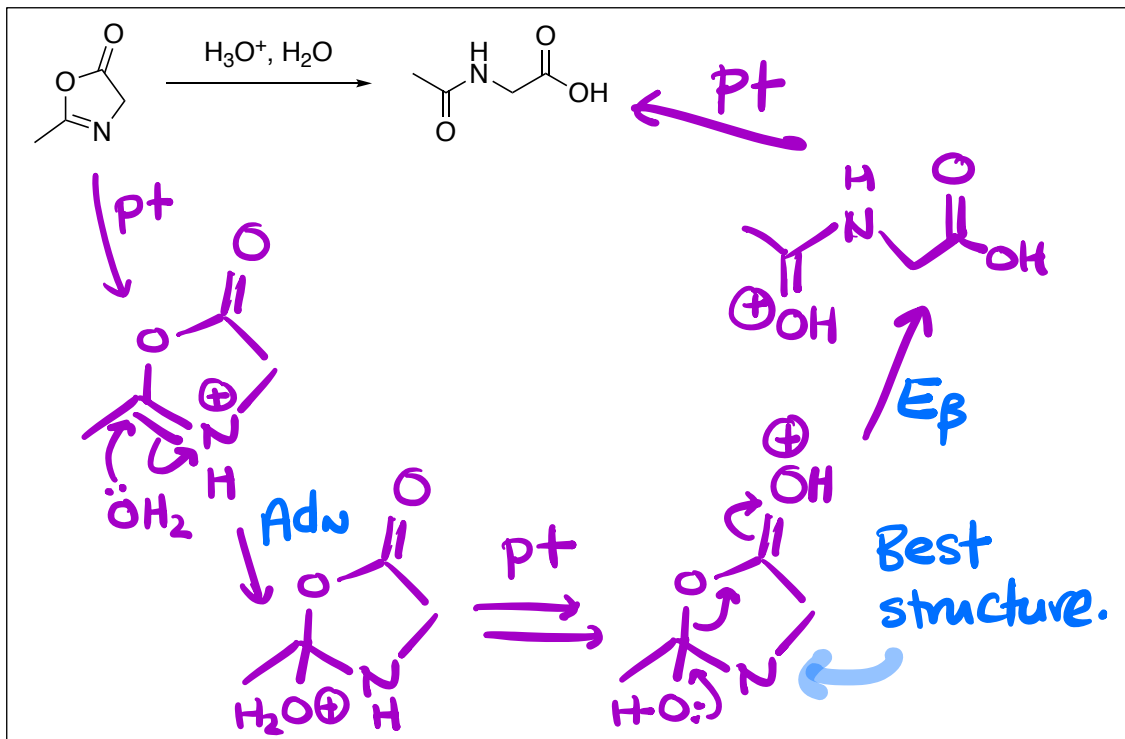


B. Draw three resonance structures for the enolate of lawsone generated when the compound is treated with lithium diisopropylamide (LDA).

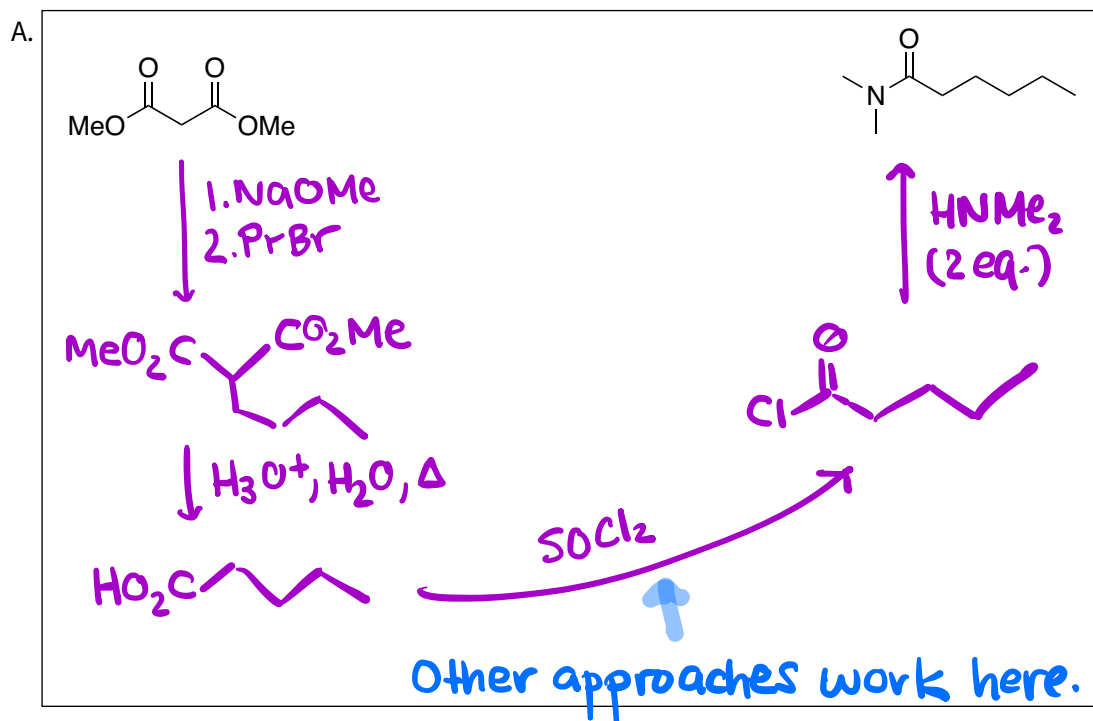


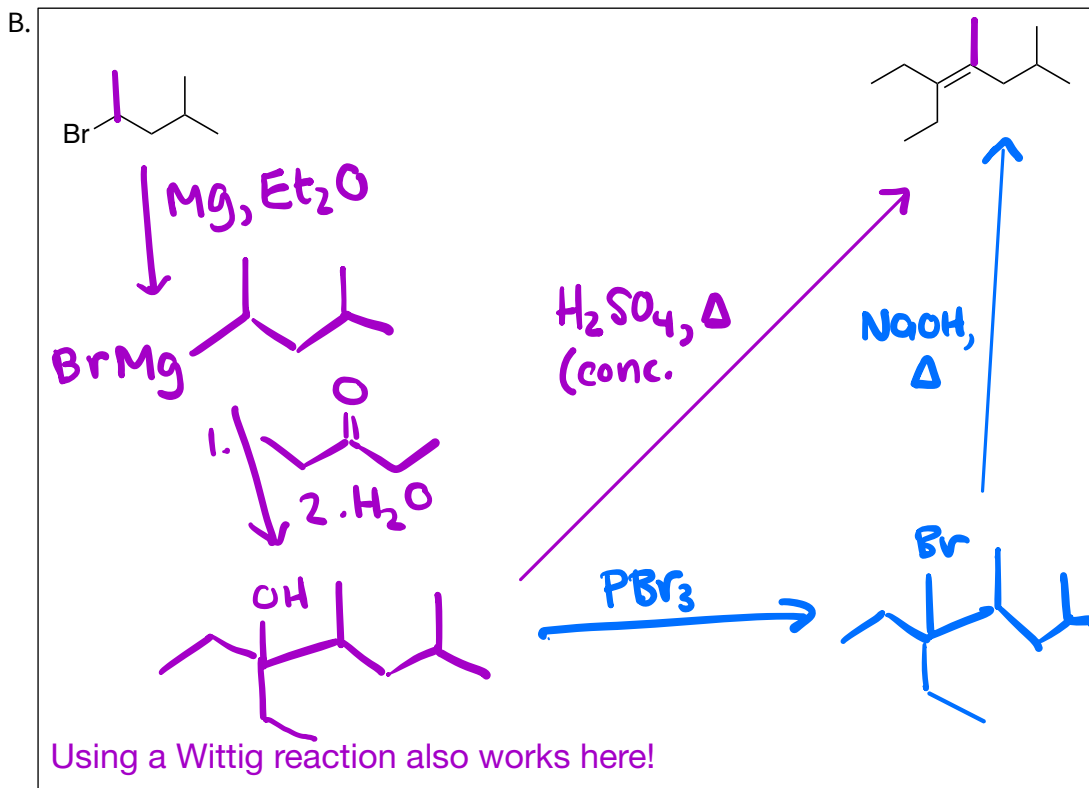
10. **10 pts.** Provide a mechanism with curved arrows and all reactive intermediates for the reaction below, *omitting proton transfers* (just write "pt" over the arrow and transfer protons as needed).

NOTE. An alternative and equally plausible mechanism involves attack by water at the ester carbonyl carbon and "opening" of the ring at the ester C–O bond, followed by amide tautomerization as needed. This is also fully correct!



11. **12 pts.** Design a synthesis of each of the following targets (right) from the indicated starting materials (left). You are welcome to use any reagents and sources of carbon, but each synthesis must start from the given starting material and must give the target in high yield.





↳ Can elaborate the alkyl halide to a ketone OR phosphonium ylide!

